

Oxidation of ethane to ethanol by nitrous oxide in a metal–organic framework with coordinatively-unsaturated iron(II) sites

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Abstract

Enzymatic heme and non-heme high-valent iron-oxo species are known to activate strong C–H bonds with relative ease, yet duplicating this reactivity in a synthetic system remains a formidable challenge. While the instability of the terminal iron-oxo moiety is perhaps the foremost obstacle, steric and electronic factors also limit the activity of previously reported mononuclear iron(IV)-oxo compounds. In particular, while nature's non-heme iron(IV)-oxos possess high-spin $S = 2$ ground states, this electronic configuration has proven difficult to achieve in a molecular species. These challenges may be mitigated within metal-organic frameworks featuring site-isolated iron centers in a constrained, weak-field ligand environment. Here, we show that the metal-organic framework $\text{Fe}_2(\text{dobdc})$ and its magnesium-diluted analogue, $\text{Fe}_{0.1}\text{Mg}_{1.9}(\text{dobdc})$, are able to convert ethane into ethanol and acetaldehyde using nitrous oxide as the terminal oxidant. Electronic structure calculations indicate that the active oxidant is likely a high-spin $S = 2$ iron(IV)-oxo species.

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Introduction

The selective and efficient conversion of light alkanes into value-added chemicals remains an outstanding challenge with tremendous economic and environmental impact,^{1,2} especially considering the recent worldwide increase in natural gas reserves.³ In nature, C–H functionalization is carried out by copper and iron metalloenzymes, which activate dioxygen and, through metal-oxo intermediates, facilitate two- or four-electron oxidations of organic substrates.^{4,5,6,7} Duplicating this impressive reactivity in synthetic systems has been the focus of intense research. In particular, iron(IV)-oxo complexes have now been structurally characterized in various geometries (octahedral, trigonal bipyramidal) and spin states ($S = 1$, $S = 2$), and have proven to be competent catalysts for a variety of oxygenation reactions.^{8,9} However, in the absence of a protective protein superstructure, terminal iron-oxo species are highly susceptible to a variety of decomposition pathways, including dimerization to form oxo-bridged diiron complexes, intramolecular ligand oxidation, and solvent oxidation.¹⁰ Tethering a molecular iron species to a porous solid support such as silica or polystyrene could potentially prevent many of these side-reactions. In practice, however, complexes heterogenized in this manner are challenging to characterize by available techniques, and additional problems associated with steric crowding, site inaccessibility, and metal leaching inevitably arise.^{11,12} Iron cations can also be incorporated into zeolites, either as part of the framework or at extraframework sites, producing reactive iron centers that have no direct molecular analogue. Fe-ZSM-5, for example, has been shown to oxidize methane to methanol stoichiometrically when pretreated with nitrous oxide.¹³ However,

characterization of these materials is nontrivial due to the presence of multiple iron species, and the nature of the active sites in Fe-ZSM-5 remains largely a matter of speculation.¹⁴

The use of a metal-organic framework to support isolated terminal iron-oxo moieties is a currently unexplored yet highly promising area of research. The high surface area, permanent porosity, chemical and thermal stability, and synthetic tunability displayed by many of these materials makes them appealing in this regard. Additionally, metal-organic frameworks are typically highly crystalline with well-defined metal centers suited for characterization by single crystal and/or powder diffraction techniques. Furthermore, while molecular iron(IV)-oxo complexes generally utilize nitrogen-based chelating ligands, the metal cations in metal-organic frameworks are often ligated by weaker-field ligands, such as carboxylates and aryloxides, which are constrained in their coordination position by the extended framework structure. Thus, in addition to increased stability, terminal oxos in these materials might also have novel electronic properties and reactivity imparted by their unique coordination environment.

Herein, we show that the high-spin iron(II) centers within $\text{Fe}_2(\text{dobdc})$ ($\text{dobdc}^{4-} = 2,5\text{-dioxido-1,4-benzenedicarboxylate}$) can activate N_2O , most likely forming a transient, high-spin iron(IV)-oxo intermediate, which rapidly reacts to afford $\text{Fe}_2(\text{OH})_2(\text{dobdc})$. Significantly, the magnesium-diluted analogue, $\text{Fe}_{0.1}\text{Mg}_{1.9}(\text{dobdc})$, is found to selectively oxidize ethane to ethanol in the presence of N_2O under mild conditions.

Results and Discussion

Of the three-dimensional iron(II)-containing metal-organic frameworks shown to be stable to desolvation,^{15,16,17,18} few possess coordinatively-unsaturated metal centers in a single, well-defined environment. The compound Fe₂(dobdc) (**1**), also known as Fe-MOF-74 or CPO-27-Fe, is rare in this regard, as the hexagonal channels of the framework are lined with a single type of square pyramidal iron(II) site (see Figure 1a). The high density and redox-active nature of these open metal sites engender excellent O₂/N₂ and hydrocarbon separation properties.^{18,19} However, with respect to the reactivity of the framework, only the hydroxylation of benzene to phenol and the oxidation of methanol to formaldehyde have been reported.^{20,21} Thus, we embarked on a study of its reactivity towards nitrous oxide, a gaseous two-electron oxidant and O-atom transfer agent that is widely employed in industry, anticipating the generation of a highly reactive iron(IV)-oxo species capable of oxidizing strong C–H bonds.

We first investigated the binding of nitrous oxide to **1** under conditions in which the Fe–N₂O interaction is reversible. Experimental studies on the coordination chemistry of N₂O are scarce, as metal–N₂O adducts are challenging to synthesize due to the poor σ -donating and π -accepting properties of the molecule.²² Indeed, of the several proposed binding modes, only one—end-on, η^1 -N—has been structurally characterized in a molecular complex.²³ To establish the coordination mode of N₂O in **1**, powder neutron diffraction data, which are very sensitive to the atomic assignment of O and N, were collected on a sample dosed with various loadings of N₂O. At low loadings, the best fit was an average of approximately 60% η^1 -O and 40% η^1 -N coordination, with Fe–N₂O distances of 2.42(3) and 2.39(3) Å, respectively. In both cases, a bent Fe–N₂O angle close

to 120° is observed (see Figure 1b). Density functional theory (DFT) studies of N₂O-bound **1** using the M06 functional²⁴ show excellent agreement with experiment (see Figure S20). Furthermore, these calculations predict the η^1 -O coordination mode to be favored over the η^1 -N mode by just 1.1 kJ/mol (see Tables S17 and S22). This is consistent with the nearly equal population split observed, although the magnitude of the difference is smaller than the reliability of the calculations.

While η^1 -O coordination with a bent Fe–O–N angle has been proposed in a variety of systems ranging from isolated metal atoms to iron zeolites,^{22,25,26} η^1 -N coordination with a bent Fe–N–N angle is much more unusual. It suggests little π -back-bonding from the metal d-orbitals into the π^* of N₂O, in contrast to previously reported vanadium and ruthenium–N₂O adducts, which have linear metal–N–N–O geometries and for which π -interactions have been invoked as significant contributors to the stability of the complexes.^{23,27,28,29} The bent geometry, long Fe–N₂O bond length, and mixed N- and O-coordination indicate N₂O is bound only weakly to the iron(II) centers in the framework, a hypothesis corroborated by *in-situ* transmission-mode infrared spectroscopy. Spectra collected on a thin-film of **1** dosed at room temperature with N₂O display a maximum at 2226 cm⁻¹, which is very close to the fundamental ν (N–N) transition for unbound N₂O (2224 cm⁻¹), suggesting a physically adsorbed phase with little to no perturbation of the N₂O molecule (see Figure S1). As expected, this interaction is fully reversible, and the band completely disappears under applied vacuum. Consistent with these experimental results, DFT studies calculate binding energies of 45.6 and 44.5 kJ/mol for the η^1 -O and η^1 -N modes, respectively, with a natural bond order analysis³⁰ showing weak back-bonding in both configurations (see Table S23).

Upon heating the N₂O-dosed framework to 60 °C, the material undergoes a drastic color change from bright green to dark red-brown that is suggestive of oxidization. In addition, *in situ* infrared studies using CO as a probe molecule show that the open metal sites, which coordinate CO strongly, have been almost entirely consumed (see Figure S9). Characterization of the resulting product is consistent with the formulation Fe₂(OH)₂(dobdc) (**2**), in which each iron center is in the +3 oxidation state and bound to a terminal hydroxide anion (see Figure 2a). Compound **2** is likely formed via a fleeting iron-oxo intermediate, which rapidly undergoes H-atom abstraction, although the source of the H-atom has not been determined. Mössbauer spectroscopy was used to probe the local environment of the iron centers in the oxidized material. The ⁵⁷Fe Mössbauer spectrum of **2** consists of a doublet characterized by an isomer shift (δ) of 0.40(2) mm/s and a quadrupole splitting ($|\Delta E_Q|$) of 0.96(1) mm/s (see Figure 2b). The isomer shift for the iron centers in **2** is similar to the parameters obtained for the peroxide-coordinated iron(III) centers in Fe₂(O₂)(dobdc),¹⁸ and is consistent with other high-spin heme and nonheme iron(III) species.^{31,32,33} In addition, the infrared spectrum of **2** shows the appearance of two new bands as compared to the unoxidized framework, which we assign as Fe–OH (667 cm⁻¹) and O–H (3678 cm⁻¹) vibrations. These bands shift to 639 and 3668 cm⁻¹, respectively, when N₂¹⁸O is employed for the oxidation; the observed differences of 28 and 10 cm⁻¹ are very close to the theoretical isotopic shifts of 27 and 12 cm⁻¹ predicted by a simple harmonic oscillator model (see Figure 3a). Partial oxidation of the framework is achieved by heating at 35 °C for 12 h, leading to the formation of Fe₂(OH)_{0.6}(dobdc) (**2'**), which has a similar infrared spectrum (though the bands associated with Fe–OH are less intense) and Mössbauer parameters (see Table S9).

The framework maintains both crystallinity and porosity after oxidation, with a Brunauer-Emmett-Teller (BET) surface area of 1013 m²/g and a Langmuir surface area of 1171 m²/g. Rietveld analysis of powder X-ray diffraction data collected at 100 K on **2** firmly establishes the presence of a new Fe–O bond, but does not reveal whether a hydrogen atom is present. However, the Fe–OH bond distance of 1.92(1) Å is consistent with the bond lengths of previously reported octahedral iron(III)-hydroxide complexes (1.84–1.93 Å) (see Figure 3b).³⁴ In addition, the *trans* Fe–O_{axial} bond is slightly elongated (Fe–O_{axial} = 2.20(1) Å; average Fe–O_{equatorial} = 2.04(1) Å), with the iron center shifted slightly out of the plane of the four equatorial oxygen atoms by 0.23(1) Å. EXAFS analysis of the same sample, as well as periodic DFT calculations, provide bond lengths that are consistent with those obtained from the diffraction data (see Table S8).

Surprisingly, the iron(III)-hydroxide species is capable of activating weak C–H bonds. When the partially oxidized sample **2'** is exposed to 1,4-cyclohexadiene (C–H bond dissociation energy of 305 kJ/mol)³⁵ at room temperature, benzene is produced as the sole product in quantitative yield. In the process, the framework converts entirely back to iron(II), as determined by Mössbauer spectroscopy. Such reactivity is rare but not unprecedented for iron(III)-hydroxide compounds. For instance, lipoxygenase, an enzyme that converts 1,4-dienes into alkyl hydroperoxides, is believed to proceed through a non-heme ferric hydroxide intermediate,⁵ and several molecular lipoxygenase mimics have also been reported to activate the C–H bond of 1,4-cyclohexadiene and other 1,4-dienes.^{36,37} However, the oxidizing power of **2** and **2'** is limited, and no reaction is observed with less activated C–H bonds.

Because the isolation of an iron(III)-hydroxide product from a reaction employing a two-electron oxidant strongly suggests the intermediacy of an iron(IV)-oxo species, we next carried out the oxidation in the presence of a hydrocarbon substrate containing stronger C–H bonds, specifically ethane (C–H bond dissociation energy of 423 kJ/mol), hoping to intercept the oxo species before its decay. Indeed, flowing an N₂O:ethane:Ar mixture (10:25:65) over the framework at 75 °C led to the formation of various ethane-derived oxygenates, including ethanol, acetaldehyde, diethyl ether, and other ether oligomers, as determined by ¹H NMR spectroscopy of the extracted products. The formation of ether products is not unprecedented, as N₂O-treated Fe-ZSM-5 forms a small amount of dimethylether in addition to methanol when exposed to methane, via a mechanism proposed to involve methyl radicals as well as multiple iron sites.³⁸ We hypothesized that the complex mixture of products was related to the close proximity of reactive iron centers, which are 8.75(2) Å and 6.84(1) Å apart across and along a channel, respectively, in **1**. To avoid oligomerization and over-oxidation, a mixed-metal metal-organic framework, Fe_{0.1}Mg_{1.9}(dobdc) (**3**), in which the iron(II) sites are diluted with redox-inactive magnesium(II) centers, was synthesized. The BET surface area of 1670 m²/g for this material falls between the surface areas of the pure iron and pure magnesium frameworks (1360 and 1800 m²/g, respectively). While determining the exact distribution of metal centers in heterometallic metal-organic frameworks is challenging, the unit cell parameters of **3** are also in between those of Fe₂(dobdc) and Mg₂(dobdc) (see Table S10), suggesting the formation of a solid solution rather than a mixture of two separate phases. Additionally, the Mössbauer spectrum of **3** shows sharp doublets with a significantly different quadrupole splitting than the all-iron analogue (2.25(1) mm/s versus 2.02(1)

mm/s in $\text{Fe}_2(\text{dobdc})$; see Table S9), indicating that the iron centers in the magnesium-diluted framework are in an altered, but uniform, environment. Thus, **3** is likely best described as containing either isolated iron centers or short multiiron segments dispersed evenly throughout a magnesium-based framework.

Exposure of **3** to N_2O and ethane under the same flow-through conditions yields the exclusive formation of ethanol and acetaldehyde in a 10:1 ratio, albeit in low yield (60% with respect to iron). Gas chromatography analysis of the headspace reveals no ethanol, acetaldehyde, or CO, suggesting the products remain bound to the framework (either at open iron or open magnesium sites), which likely explains the high ethanol selectivity. While the framework is still highly crystalline after N_2O /ethane treatment, Mössbauer spectroscopy reveals that roughly 90% of the iron centers have decayed into a species with similar spectral parameters as **2** (see Figure S18 and Table S9). We propose that formation of iron(III)-hydroxide or alkoxide decay products prematurely halts the catalytic cycle, which leads to substoichiometric yields of hydroxylated product (see Figure 4). Because glass can be a source of H-atoms, the reaction was subsequently repeated in a batch, rather than flow-through mode, in a Teflon-lined stainless-steel bomb, which produced both higher yields with respect to iron (turnover number = 1.6) and selectivities (25:1 ethanol:acetaldehyde), showing that the system can indeed be modestly catalytic if competing substrates are excluded. However, the yield based on ethane (roughly 1%) is still too low for practical purposes.

As the high reactivity of the iron-oxo species precluded isolation in both $\text{Fe}_2(\text{dobdc})$ and its magnesium-diluted analog, electronic structure calculations were performed on $\text{Fe}_2(\text{O})_2(\text{dobdc})$ (**4**) to gain insight into the geometric and electronic

structure of iron-oxo units supported within the framework. First, periodic PBE+U^{39,40} geometry optimizations were performed on **4** for the singlet, triplet, and quintet spin states. A quintet ground state was predicted, with a short Fe–O bond length of 1.64 Å, consistent with that of previously reported iron(IV)-oxo complexes (see Figure 5 and Table S11).⁸ The periodic structure was then truncated to an 89-atom model cluster^{41,42} containing three metal centers, six organic linkers, and an oxo moiety to facilitate calculations using more accurate methods. The cluster calculations were simplified by replacing the two peripheral iron(II) centers with closed-shell zinc(II) centers, which have the same charge and a similar ionic radius to iron(II) and magnesium(II) cations (see Figure S19). The geometry of this cluster was then optimized for the ground state, with all atoms except for the central iron and its first coordination sphere frozen at the coordinates from the periodic PBE+U optimization. As shown in Table 1, the M06²⁴ calculations also predict a quintet ground state. Further calculations were performed with several other exchange-correlation functionals, and in each case the ground state was found to be a quintet (see Tables S11–S16). Note that similar results were obtained when the Zn(II) centers in the 89-atom cluster were replaced with Mg(II) centers (see Tables S20 and S21).

The electronic structure of the cluster model of **4** was further examined with single-point multiconfigurational complete active space (CASSCF) calculations followed by second-order perturbation theory (CASPT2).^{43,44} Again, the ground state is predicted to be the quintet state (see Table 1 and Table S18). Both M06 and CASPT2 yield a spin density of ~3.7 on iron, consistent with four unpaired spins mainly localized on the metal (see Tables S13 and S19). Density functional and CASPT2 calculations were also

performed on the cluster model of **2**; all calculations led to a high-spin sextet ground state for the iron(III) center (see Tables S11–S16 and Table S18).

Table 1 | Calculated relative energies (kJ/mol) of the cluster model of **4.**

<i>S</i>	M06	CASPT2
0	210.6	249.4
1	136.4	127.6
2	0.0	0.0

While spectroscopic and theoretical studies have long attributed the reactivity of non-heme enzymatic and synthetic iron(IV)-oxo complexes to a quintet spin state,⁴⁵ only a small handful of mononuclear high-spin iron(IV)-oxo species have been characterized,^{46,47,48,49} with all but one exhibiting a trigonal bipyramidal coordination geometry.⁵⁰ In these systems, the oxo moiety is either extremely unstable—[Fe(O)(H₂O)₅]²⁺, for example, has a half life of roughly 10 s—or inaccessible to substrates due to bulky ligand scaffolds, leading to sluggish reactivity. On the other hand, the Fe₂(dobdc) framework features sterically accessible, site-isolated metal centers entrenched in a weak-field ligand environment. Utilizing these two properties, it is possible not only to generate such a species, albeit fleetingly, but also to direct it towards the facile activation of one of the strongest C–H bonds known.

Concluding remarks

The foregoing results demonstrate through reactivity studies, detailed characterization of decay products, and theoretical calculations that the iron-based metal-organic frameworks Fe₂(dobdc) and Fe_{0.1}Mg_{1.9}(dobdc) are very likely capable of

supporting fleeting iron(IV)-oxo species possessing an unusual $S = 2$ spin state. With this, $\text{Fe}_2(\text{dobdc})$ has now been shown to stabilize iron-superoxo, peroxy, hydroxo, as well as oxo intermediates, highlighting the promise of metal-organic frameworks both as catalysts and as scaffolds for interrogating reactive metal species. Future work will focus on: (i) further exploring the reactivity of $\text{Fe}_2(\text{dobdc})$ and its expanded analogues with ethane and other hydrocarbon substrates, as well as continued efforts to isolate the iron-oxo species, (ii) the use of dioxygen as the terminal oxidant in such systems, and (iii) the design, synthesis, and reactivity of other metal-organic frameworks with coordinatively-unsaturated iron sites.

Methods Summary

Synthesis of $\text{Fe}_2(\text{OH})_{0.6}(\text{dobdc})$ (2') and $\text{Fe}_2(\text{OH})_2(\text{dobdc})$ (2)

An evacuated Schlenk flask containing fully desolvated $\text{Fe}_2(\text{dobdc})$ (100 mg, 0.33 mmol) was placed under an atmosphere of 30% N_2O and 70% N_2 . The flask was immersed in an oil bath, and the temperature was increased by 10 °C every 12 h, from 25 °C up to 60 °C, to obtain $\text{Fe}_2(\text{OH})_2(\text{dobdc})$ as a dark red-brown solid. When the reaction was stopped after 12 h at 35 °C, the partially oxidized $\text{Fe}_2(\text{OH})_{0.6}(\text{dobdc})$ (as determined by Mössbauer spectroscopy) was obtained. Anal. Calc. for $\text{C}_8\text{H}_4\text{Fe}_2\text{O}_8$: C, 28.28; H, 1.19. Found: C, 29.18; H, 1.16. IR (solid-ATR): 3679 (m), 1532 (s), 1450 (s), 1411 (s), 1361 (s), 1261 (s), 1154 (w), 1129 (w), 1077 (w), 909 (m), 889 (s), 818 (s), 807 (s), 667 (s), 630 (m), 594 (s), 507 (s).

Synthesis of $\text{Fe}_{0.1}\text{Mg}_{1.9}(\text{dobdc})$ (3)

In a 500-mL Schlenk flask, $\text{H}_4(\text{dobdc})$ (1.8 g, 8.8 mmol), MgCl_2 (1.5 g, 15 mmol), and FeCl_2 (0.84 g, 6.6 mmol) were dissolved in a mixture of 310 mL of DMF and 40 mL of methanol. The reaction was stirred vigorously at 120 °C for 16 h. The precipitate was filtered and stirred with 250 mL of fresh DMF at 120 °C for 3 h. Two more DMF washes at 120 °C were performed, after which the precipitate was filtered and soaked in methanol at 60 °C. The methanol exchanges were repeated until no DMF stretches were apparent in the infrared spectrum. The framework was fully desolvated under dynamic vacuum (<15 μbar) at 210 °C for 2 days to afford $\text{Fe}_{0.1}\text{Mg}_{1.9}(\text{dobdc})$ as a bright yellow-green solid (2.0 g, 8.2 mmol, 93% yield). The iron to magnesium ratio was determined by ICP-OES. Anal. Calc. for $\text{C}_8\text{H}_2\text{Fe}_{0.1}\text{Mg}_{1.9}\text{O}_6$: C, 39.08; H, 0.82. Found: C, 39.37; H, 0.43.

IR (solid-ATR): 1577 (s), 1484 (m), 1444 (s), 1429 (s), 1372 (s), 1236 (s), 1210 (s), 1123 (m), 911 (m), 892 (s), 828 (s), 820 (s), 631 (s), 584 (s), 492 (s).

Reactivity of Fe₂(dobdc) (1) and Fe_{0.1}Mg_{1.9}(dobdc) (3) with N₂O and C₂H₆

In a typical flow-through experiment, a mixture of gases (2 mL/min N₂O, 10 mL/min C₂H₆, and 8 mL/min Ar for a total flow 20 mL/min) was flowed over a packed bed of metal-organic framework (50 to 100 mg) contained within a glass column. The column was heated to 75 °C for 24 h, after which the products were extracted with CD₃CN (3 × 1 mL) and analyzed by ¹H NMR using 1,4-dichlorobenzene as an internal standard. While a cold bath maintained at −78 °C was installed downstream of the glass reactor in order to collect condensable organic products, at the temperatures tested all the products appeared to remain bound to the framework.

In a typical batch experiment, a Parr bomb was charged with N₂O (1.5 bar) and C₂H₆ (7.5 bar) and heated to 75 °C in a sand bath. After 24 h, the bomb was cooled and the products extracted with CD₃CN.

Electronic structure calculations

The structures of **2** and **4** were optimized using periodic boundary conditions and the PBE+U exchange-correlation functional. From each of these structures, we carved out a model cluster containing three iron centers along a single helical chain and six organic linkers. These clusters are analogous to the 88-atom cluster model of Fe₂(dobdc) employed previously.⁴¹ The cluster models were further simplified by substituting the two peripheral iron(II) centers with zinc(II) centers, while keeping only the central iron(II) in the cluster. Constrained geometry optimizations were performed where only the central iron and the six oxygen atoms (plus the hydroxide hydrogen in compound **2**) of its first coordination sphere were allowed to relax. Single-point multiconfigurational complete active space (CASSCF) calculations followed by second-order perturbation theory (CASPT2) were performed at PBE-optimized (PBE/SDD(Fe,Zn),6-31G(d)(C, H, O)) geometries of the cluster models of **2** and **4**.

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D. J. X., E. D. B., and J. R. L. planned and executed the synthesis, characterization, and reactivity studies. J. A. M., W. L. Q., M. R. H., and C. M. B. analyzed the powder neutron and X-ray diffraction data. N. P. and P. V. performed the cluster DFT calculations. J. B. and K. L. performed the periodic DFT calculations. A. L. D. performed the CASSCF/PT2 calculations. D. G. T. and L. G. conceived and managed the computational efforts. F. B., V. C., and S. B. carried out the *in situ* transmission FTIR studies, and J. Y. supervised EXAFS analysis. All authors participated in the preparation of the manuscript.

Additional information

Further characterization, experimental, and computational details are described in the supplementary material available online at www.nature.com/naturechemistry (Tables S1-S19 and Figures S1-S20). Crystallographic data for Fe₂(dobdc) loaded with 0.35, 0.6, and 1.25 equivalents of N₂O and Fe₂(OH)₂(dobdc) (data collected at 100 and 298 K) have been deposited in the Cambridge Structural Database as CCDC 977092-977096.

Competing financial interests

The authors declare no competing financial interests.

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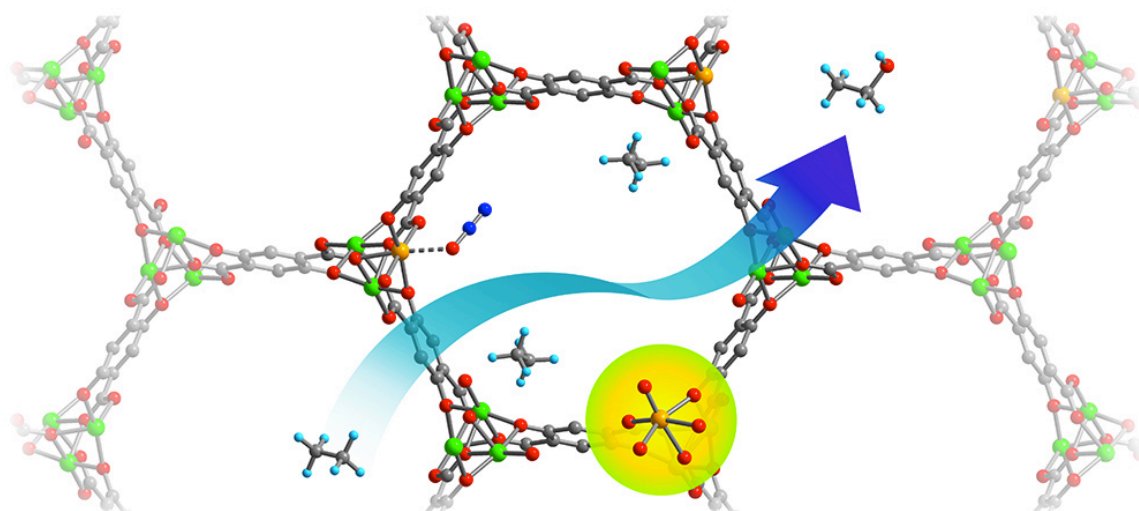
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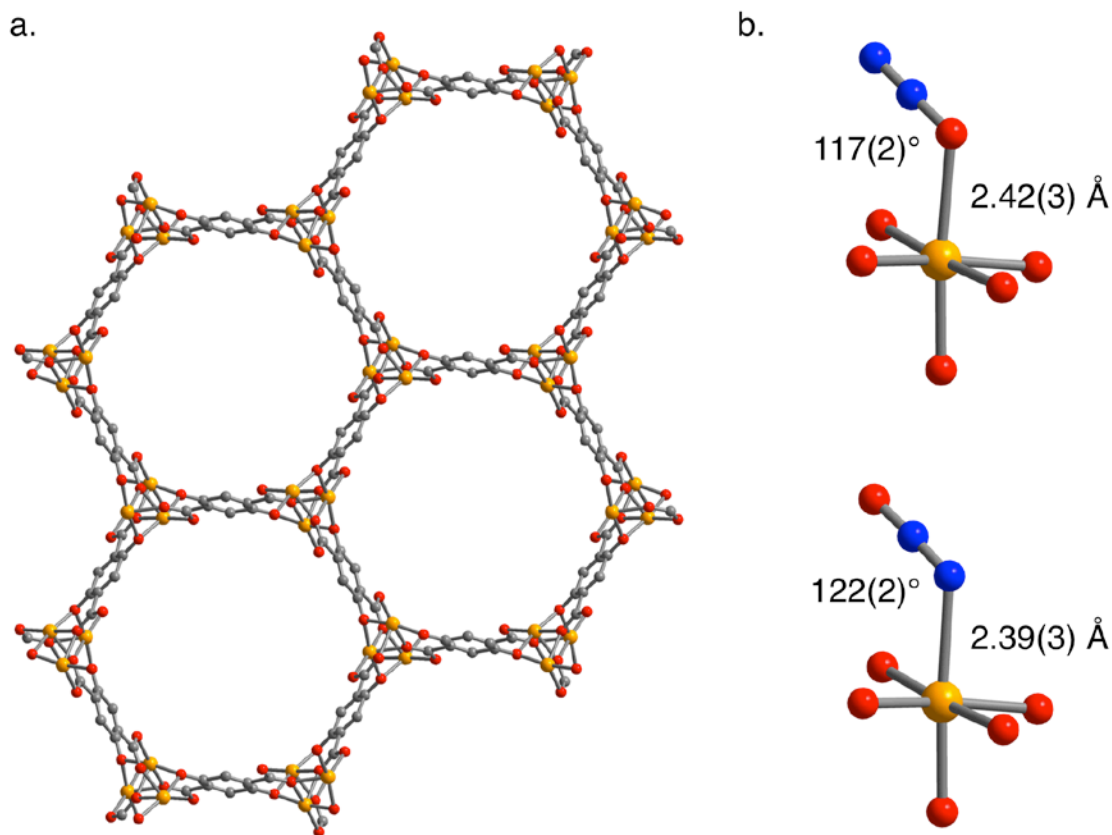


Figure 1 | Structure of bare and N_2O -dosed $\text{Fe}_2(\text{dobdc})$. **a.** Structure of $\text{Fe}_2(\text{dobdc})$, showing hexagonal channels lined with 5-coordinate iron(II) sites. The view is down the c axis, along the helical chains of iron(II) ions. **b.** Experimental structures for N_2O binding in $\text{Fe}_2(\text{dobdc})$ loaded with 0.35 equivalents of N_2O at room temperature and then slowly cooled to 10 K. The molecule binds with a bent $\text{Fe}-\text{N}_2\text{O}$ angle, with a mixture of 60% $\eta^1\text{-O}$ coordination and 40% $\eta^1\text{-N}$ coordination. For comparison of calculated structures with experimental, see Figure S20.

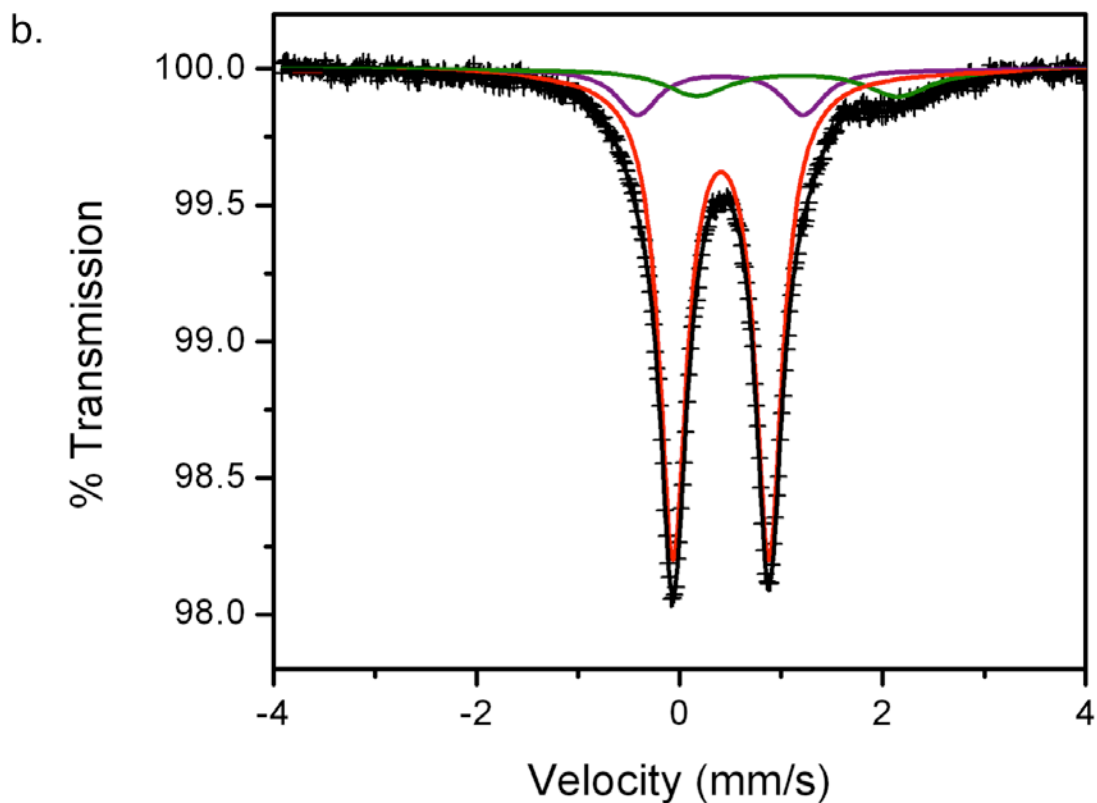
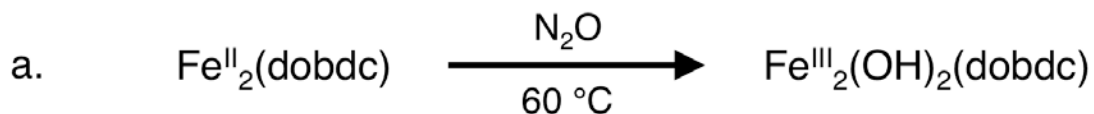


Figure 2 | Preparation and Mössbauer spectrum of $\text{Fe}_2(\text{OH})_2(\text{dobdc})$. **a.** Reaction scheme for the preparation of **2** from $\text{Fe}_2(\text{dobdc})$. **b.** Mössbauer spectrum of **2**, with the fit in black. The red component has parameters consistent with high-spin Fe(III) ($\delta = 0.40(2)$ mm/s, $|\Delta E_Q| = 0.96(1)$ mm/s, area = 80(2)%). A minor component (green) is assigned as unreacted Fe(II) sites, and another minor component (purple) is assigned as an amorphous Fe(III) decomposition product.

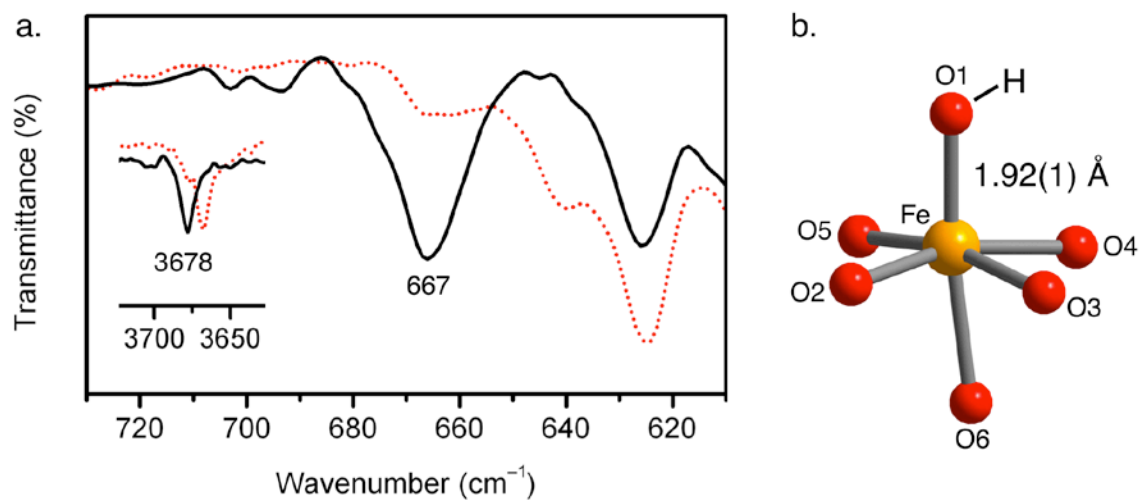


Figure 3 | Structure and infrared spectrum of Fe₂(OH)₂(dobdc). **a.** Infrared spectrum of a partially oxidized sample, Fe₂(OH)_{0.6}(dobdc) (black) and Fe₂(¹⁸OH)_{0.6}(dobdc) (dotted red). The peaks at 667 and 3678 cm⁻¹ shift to 639 and 3668 cm⁻¹, respectively, upon ¹⁸O labeling. **b.** The structure of Fe₂(OH)₂(dobdc) obtained by powder X-ray diffraction data (100 K). Selected interatomic distances (Å) for **1**: Fe–O1 = 1.92(1); Fe–O2 = 2.01(1); Fe–O3 = 2.08(1); Fe–O4 = 2.04 (1); Fe–O5 = 2.04(1); Fe–O6 = 2.20(1); Fe–Fe = 3.16(1).

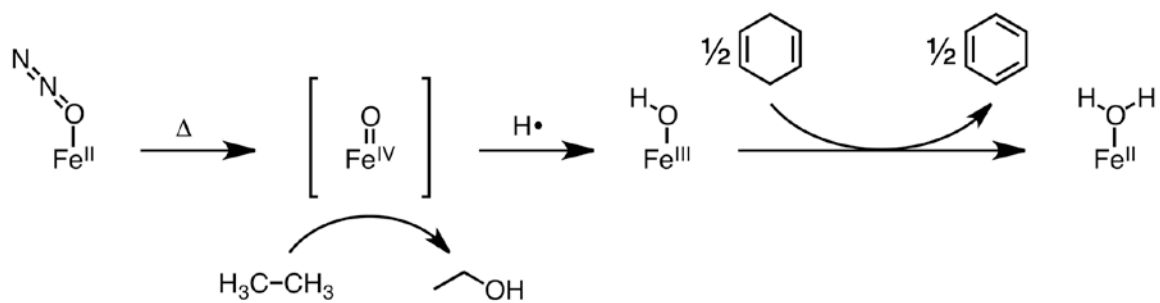


Figure 4 | N_2O activation and reactivity of $\text{Fe}_2(\text{dobdc})$.

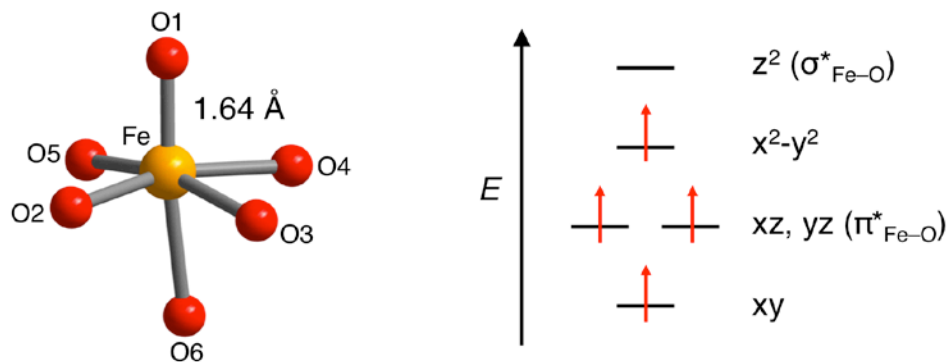


Figure 5 | Structure and qualitative MO diagram of $\text{Fe}_2(\text{O})_2(\text{dobdc})$. DFT and CASSCF/PT2 studies predict a short iron-oxo bond (1.64 Å) and a high-spin, $S = 2$ spin ground state for iron(IV)-oxos installed in the $\text{Fe}_2(\text{dobdc})$ framework. Selected interatomic distances (Å) for **1**: Fe–O1 = 1.638; Fe–O2 = 2.004; Fe–O3 = 2.127; Fe–O4 = 2.019; Fe–O5 = 2.054; Fe–O6 = 2.140.